

Day 7

1. Getting started

```
// Include JavaScript in HTML
<script src="app.js"></script>

// Write in console
console.log("Hello, JavaScript!");
```

2. Basics and syntax

```
javascript// Comments
// This is a single-line comment
/* This is a multi-line comment */

console.log("Output to console");
alert("Pop-up message");
let name = prompt("What's your name?");
```

3. Variables and data types

Feature	var	let	const
Scope	Function/Global	Block (<code>{ }</code>)	Block (<code>{ }</code>)
Reassignable	Yes	Yes	No
Redeclarable	Yes	No	No
Hoisting	Yes (initialized as undefined)	Yes (in Temporal Dead Zone)	Yes (in Temporal Dead Zone)

The **Temporal Dead Zone (TDZ)** is the period between the start of a block's scope and the actual declaration of a variable using `let` or `const`. During this time, the

variable exists in memory but is **inaccessible**; attempting to use it will result in a `ReferenceError`

```
javascript// Variables with let and const
let age = 25;
const PI = 3.14159;
var city = "Bengaluru"; // older way, avoid
```

```
// Data types
let string = "Hello";
let number = 42;
let decimal = 3.14;
let isActive = true;
let empty = null;
let notDefined = undefined;
```

```
// Type checking
console.log(typeof string); // "string"
console.log(typeof number); // "number"
```

4. Operators and expressions

```
javascript// Arithmetic
let sum = 10 + 5;    // 15
let product = 4 * 3; // 12
let power = 2 ** 3;  // 8 (exponent)

// Comparison
console.log(5 === 5); // true (strict equality)
console.log(5 == "5"); // true (loose equality)
console.log(5 !== "5"); // true (strict not equal)

// Logical
```

```
console.log(true && false); // false  
console.log(true || false); // true
```

```
// Template literals
```

```
let name = "Santhosh";
```

```
console.log(`Hello, ${name}!`); // "Hello, Santhosh!"
```
